A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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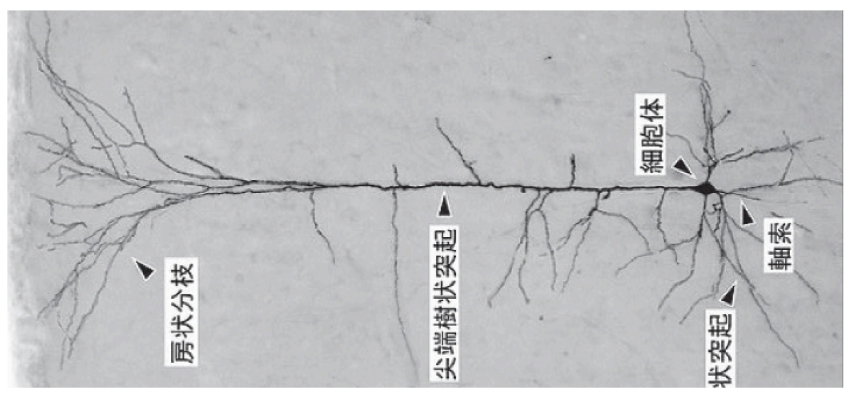
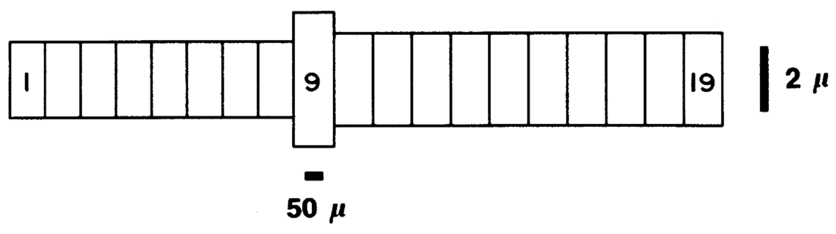
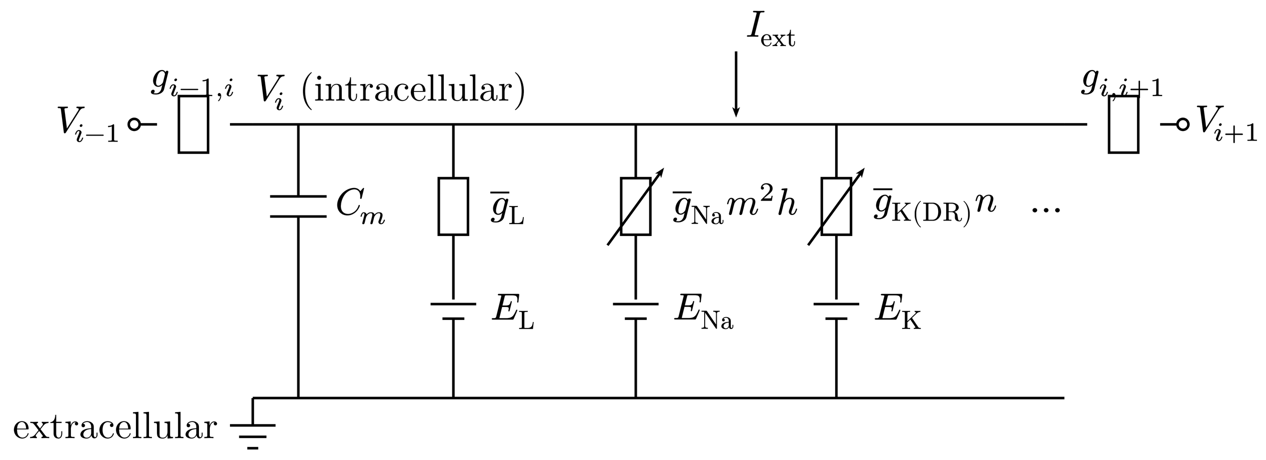
Introduction



CA3 pyramidal neuron [1]



Modelled as **19 compartments** [2]



Equivalent circuit of compartment i

Each compartment: capacitance, **variable** ionic conductances, axial coupling $g_{i,i\pm1}$.

Conductances depend on **gate variables**, ODEs with rate functions α, β:

$$I_{Na} = \bar{g}_{Na}\, m^2 h\, (V - E_{Na})$$

$$\frac{dm}{dt} = \alpha_m(V)(1-m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1-u)/(\exp((13.1-u)/4)-1)$$

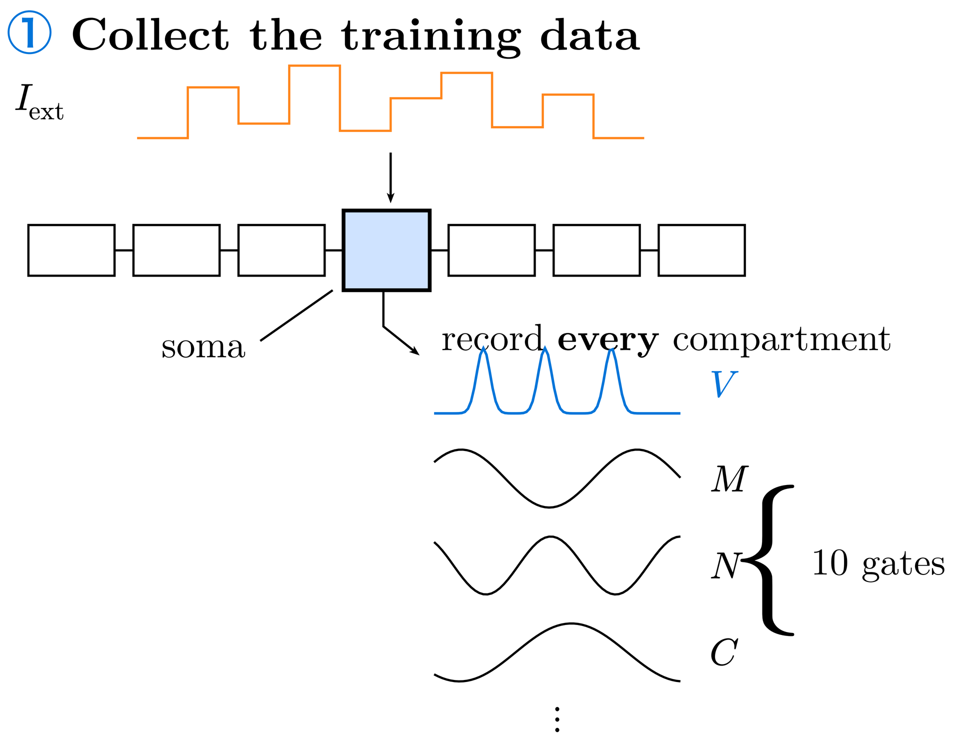
→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

Goal

Test whether **complex gate dynamics** remain tractable to a **simple regression** (SINDy) after a coordinate change — and thereby reproduce the membrane-potential response with **fewer state variables**.

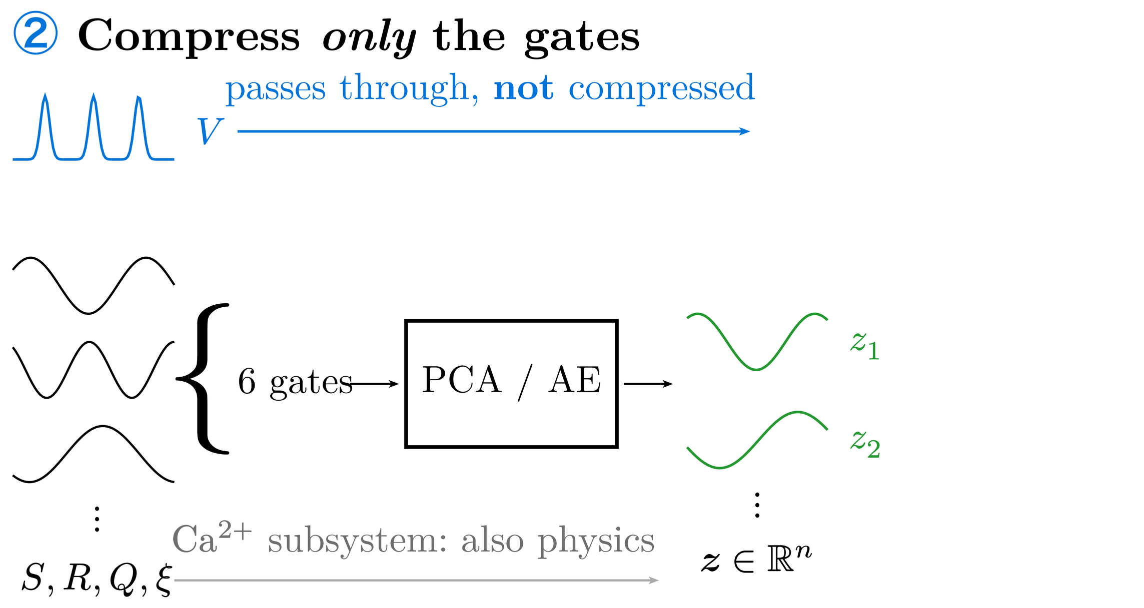
Methods

① Collect the training data



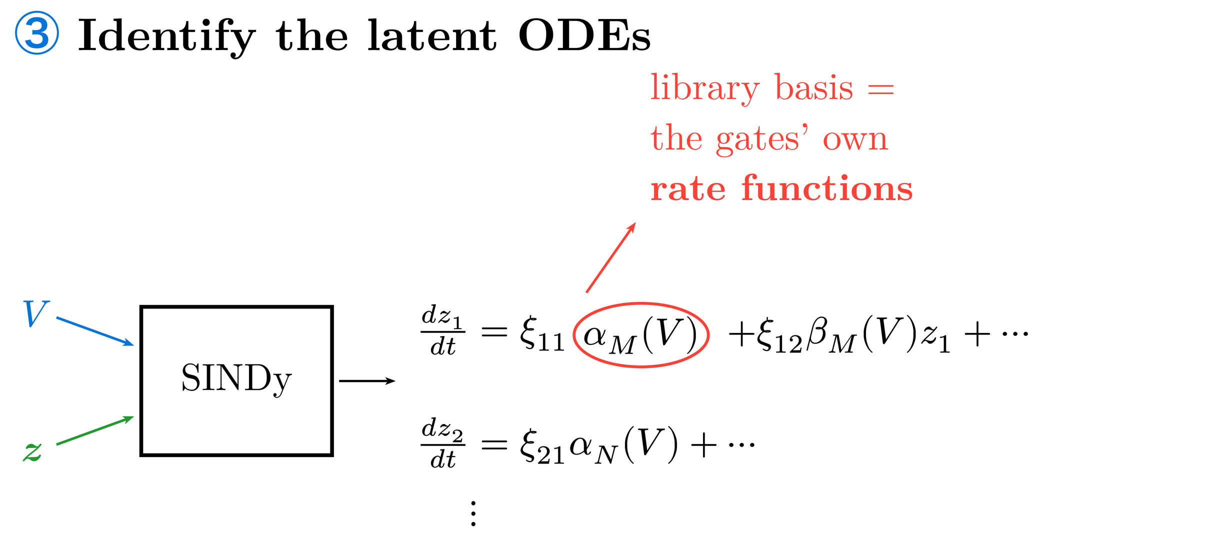
Inject a **random pulse train** at the soma; record V and **10 gates** for **all 19 compartments**.

② Compress *only* the gates



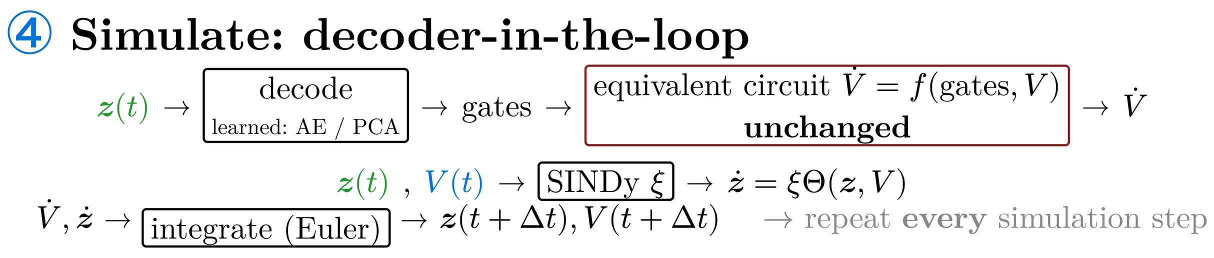
6 voltage-dependent gates ride a **low-dimensional manifold** → $\boldsymbol{z} \in \mathbb{R}^n$, $n = 5$. V and Ca^{2+} **stay uncompressed**.

③ Identify the latent ODEs



SINDy [3] fits **only** $\dot{\boldsymbol{z}}$; $\dot{V}$ and Ca^{2+} keep **original physics**. Library is **physics-informed**: gates' own $\alpha(V), \beta(V)$, not a **41-term** generic one.

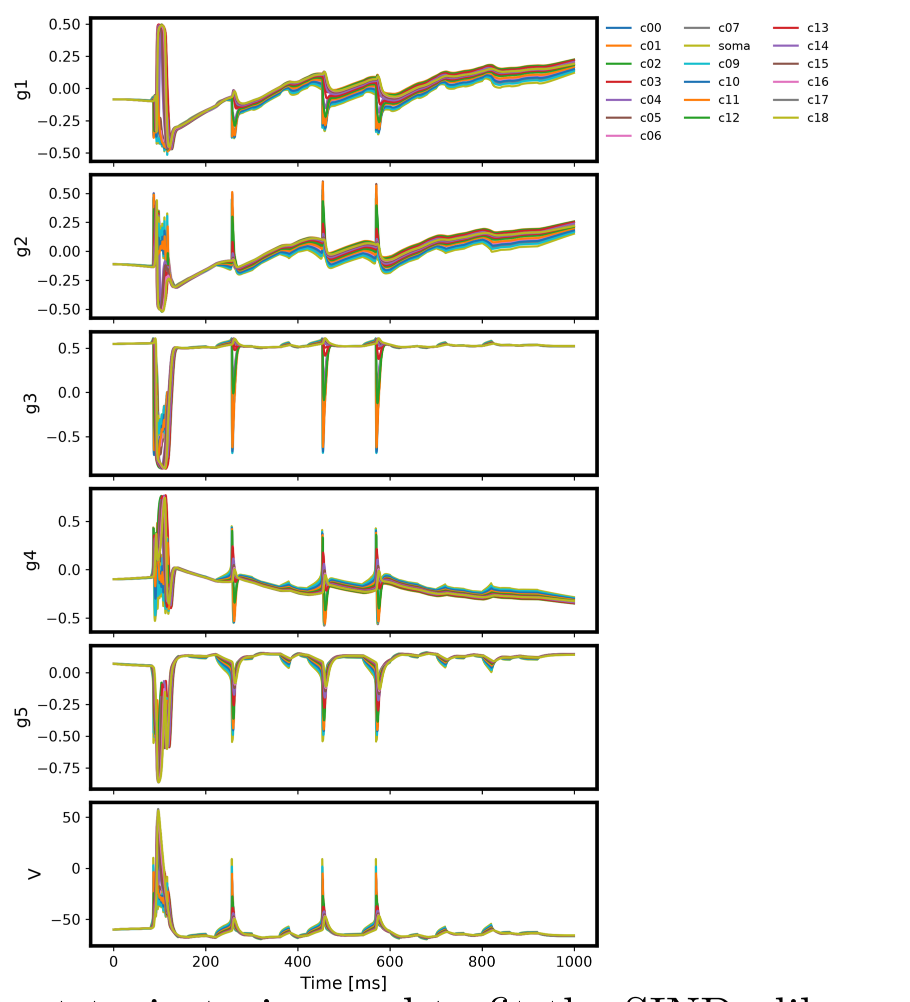
④ Simulate: decoder-in-the-loop



Each step: $\boldsymbol{z}$ → **decode** → gates feed the **unchanged** equivalent-circuit $\dot{V}$; SINDy updates $\boldsymbol{z}$ **in place of** the original gate ODEs.

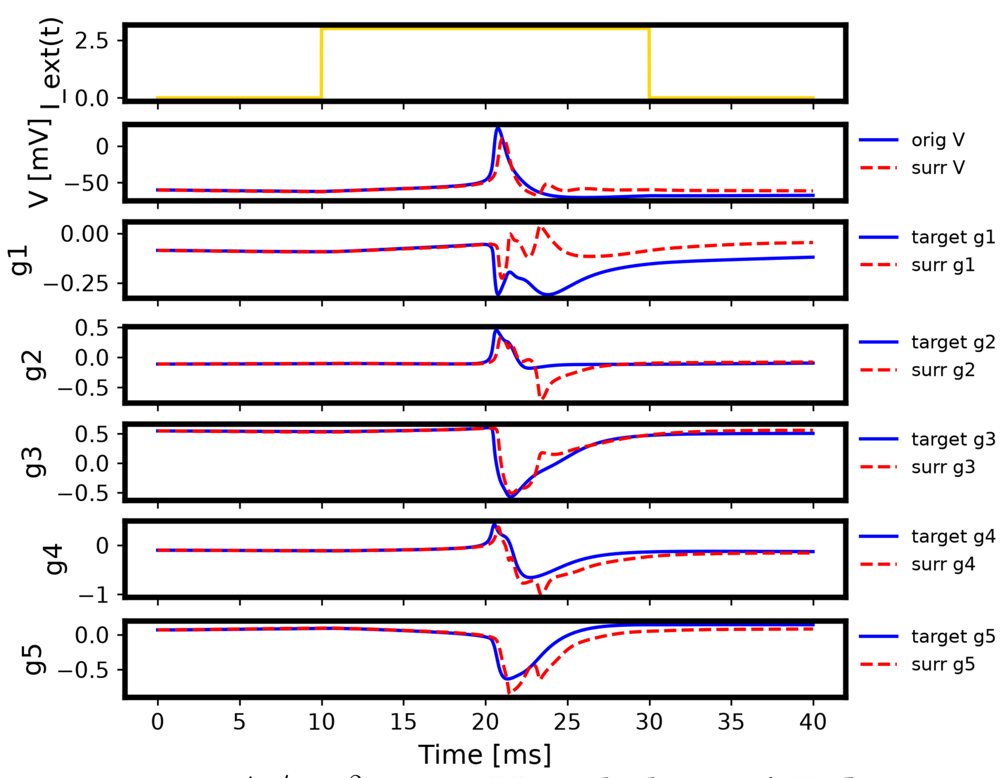
Results and Discussion

Training data (preprocessed)



Latent trajectories used to fit the SINDy library.

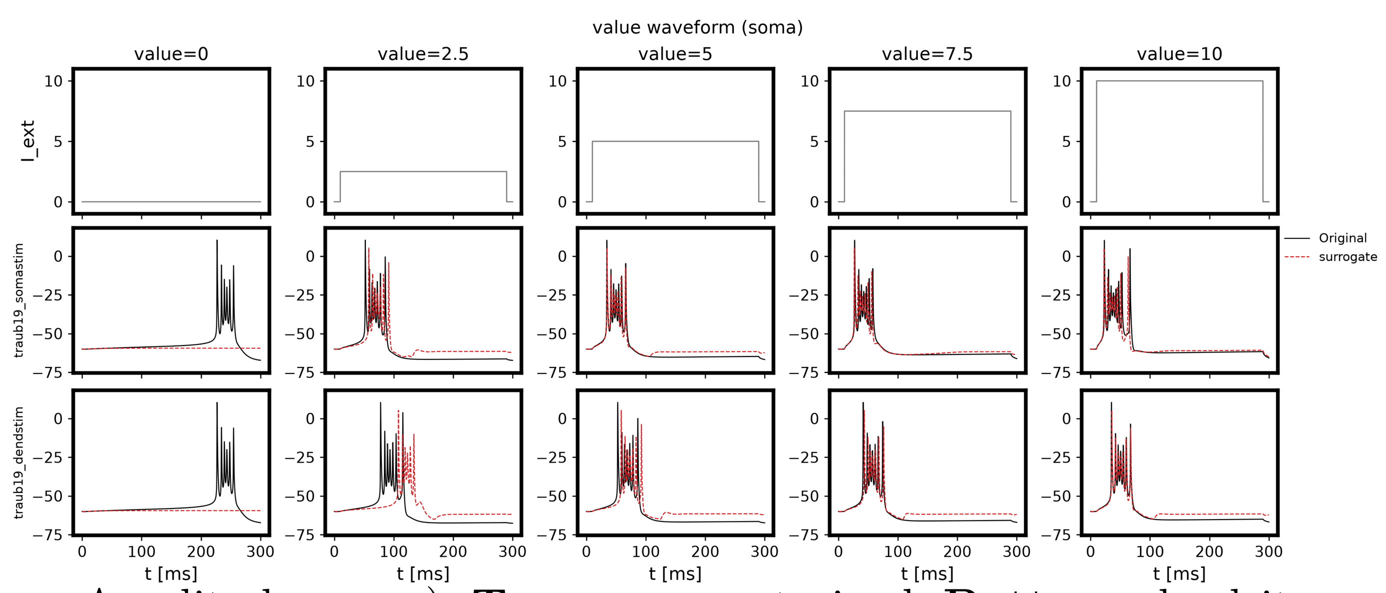
① Single action potential



20 ms, 3 μ A/cm² step: V and the **5 AE latents**.

- **When** it fires is accurate (latency err **0.3** ms, AHP timing gap **3.1 ms**).
- **How big** is systematically **underestimated** — peak **13 mV low** (amplitude gap **12.9 mV**), rise / fall rate gap **21.0** / **10.3 mV/ms** — yet AHP depth gap is only **0.28** mV.

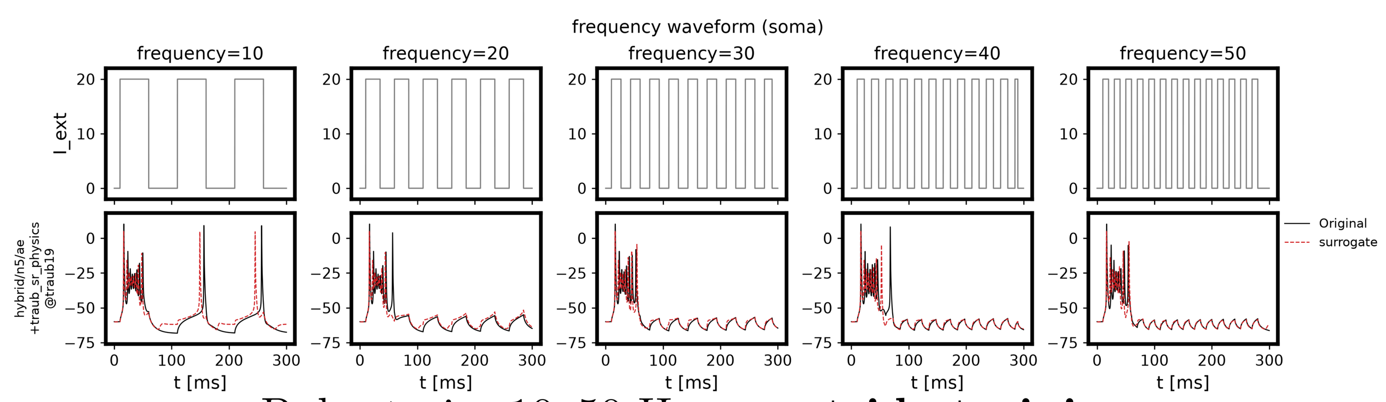
② New stimulus site



Amplitude sweep). **Top**: soma, as trained. **Bottom**: dendrite, **unseen**.

- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

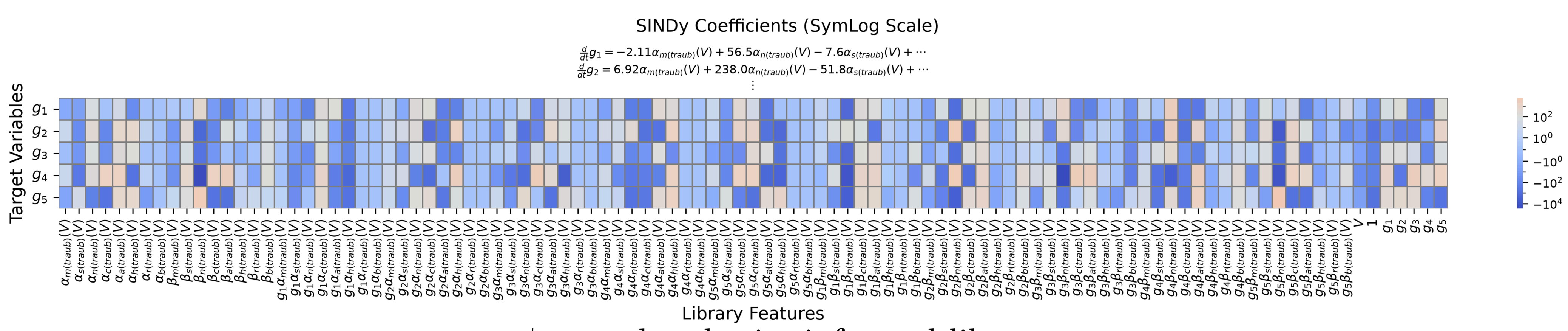
③ Unseen periodic drive



Pulse train, 10–50 Hz — **outside training**.

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

④ Identified latent equations



ξ over the physics-informed library.

- **79.6% non-zero** — accurate but **not sparse**.

Conclusion

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding (n=5)**, not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site · frequency**.
- **Core finding**: SINDy succeeds **not because we compressed** (n: 6→5) — ⑤ shows the **same result uncompressed** ($n = 6$), so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).

→ **Future**: sparsify ξ, push n down, test on other channel models.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] "錐体細胞." Accessed: Dec. 08, 2025. [Online]. Available: https://doi.org/10.14931/bsd.2091

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.